

Non-Fundamental by Impossibility: A Public-Data Bound on Panic and Friction in the March 2023 USDC De-Peg

Anonymous Submission to FC 2027

Abstract. We give a reusable, public-data instrument for a disclosed-impairment de-peg. If fundamental value is expected disclosed backing, redemption is at par under solvency, and recovery on the impaired fraction ϕ lies in $[0, 1]$, then fundamental value cannot fall below $1 - \phi$, and a Corollary extends the floor to any monotone-utility valuation. In the March 2023 USDC de-peg, SVB’s failure trapped about 8% of reserves and the coin fell to \$0.877, below the \$0.92 floor. The trough requires unremediated-loss probability $q^* = 1.54$, which exceeds one. At the disclosed ϕ , at least 35% of the de-peg is non-fundamental on the composite series, and 19–35% across feed troughs. The trough is unrationalizable for any impairment below 12.3%. Hourly evaluation dates the impossibility to nine breaching hours on 11 March. Two supporting analyses stay outside it. A global-games ex-ante fragility ordering matches the realized troughs. A specificity panel gives one true positive (USDC, 2023), one regulator-quantified, issuer-contested episode on which the bound returns no verdict (Tether, 2019), and one degenerate control (USDT, Terra episode, May 2022, $\phi = 0$).

Keywords: Stablecoins · Bank runs · Global games · De-peg · Financial stability · Market microstructure

1 Introduction

Stablecoins are becoming payment infrastructure [38], and with the GENIUS Act they now operate under their first comprehensive U.S. federal framework. That framework must cover reserve composition, redemption rights, and disclosure before the next peg breaks. A fiat-backed stablecoin that trades below par admits two readings. Under *rational repricing*, an impaired reserve is worth less. Under *non-fundamental* repricing, a self-fulfilling coordination run and microstructure friction push the price below the floor a reserve loss can justify. No measurement of March 2023 cited in Section 2 forms that floor, and the split between them should set the policy response between backstops and reserve quality.

This paper offers a reusable *estimator* that applies to any disclosed-impairment de-peg. From an issuer’s *disclosed* reserve impairment, derive a worst-case floor. Test whether the observed trough breaches that floor. Read the breach, if any, as a bound on the non-fundamental share. A supervisor can run it on public data within hours of a break. It returns no verdict when the

trough does not breach the floor. We apply it to the March 2023 de-peg of USD Coin (USDC). The failure of Silicon Valley Bank (SVB) trapped about 8% of USDC’s reserves, and the coin reached a \$0.877 trough. A joint federal backstop [5] restored the peg within days. The SVB failure was an idiosyncratic shock with a dated disclosure, and its reversal was complete. The counterfactual “no panic” price is therefore pinned twice, as a worst-case floor and, ex post, as the restored peg.

Under expected-loss pricing of the disclosed impairment, fundamental value cannot fall below a floor that a preference-free Corollary extends to any monotone-utility valuation. The observed trough breaches that floor at every admissible unremediated-loss probability and recovery rate. No rational-pricing account is consistent with the trough. The resulting non-fundamental share stays positive across impairment readings, and at the full re-peg it is 100% ex post, an identity (Section 4).

The paper makes two contributions:

- **A model-free impossibility bound** (Proposition 1): fundamental value cannot fall below $1 - \phi$, so the observed trough requires unremediated-loss probability $q^* = 1.54$, which exceeds one and is unrationalizable under expected-loss pricing.
- **An offline, keyless reproduction path**: every reported number regenerates from a frozen snapshot with no network call and no API key (Appendix C).

Two supporting analyses do not enter the estimator: a global-games exposure ordering that the realized troughs match, with null probability 1/2 (Section 5), and a specificity panel of three candidate episodes (Section 6).

2 Related Work

The theory of runs begins with Diamond and Dybvig [14], whose demand-deposit model leaves the run *probability* undetermined, and global-games methods select a unique equilibrium [7,33,19]. From that literature we take only the selection result, in which the marginal agent holds a diffuse belief over the aggregate withdrawal mass. We do not reproduce the uniqueness argument, and our result does not depend on it.

A literature since 2020 applies run theory to stablecoins, building on design taxonomies [31,24] and peg-arbitrage mechanisms [28]. Ma, Zeng, and Zhang [29] build a global-games stablecoin model, finding USDC’s arbitrage less centralized than USDT’s. Anadu et al. [3] analyze the March 2023 de-peg via a discrete “break-the-buck” threshold. Watsky et al. [42] give the institutional reason for that threshold, a closed redemption window. USDC’s primary redemption window was open only to Circle’s approved customers and was constrained by U.S. banking hours over the March 10–13 weekend. Secondary prices therefore de-pegged while the arbitrage loop was closed. A Federal Reserve post-mortem co-authored by Watsky reports the same mechanism [16]. On AMM liquidity

around the SVB failure, Hernandez Cruz et al. [23] run a difference-in-differences comparing USDC with USDT. Uhlig [37] finds many TerraUSD holders behaved as rational Bayesian updaters in the May 2022 TerraUSD collapse, which is evidence that rational and panic behavior are hard to separate without a structural restriction. Liu, Makarov, and Schoar [26] reconstruct that collapse wallet-by-wallet. Contemporaneous work surveys stablecoins as a payment technology and the financial risks they pose [2].

Other work measures the March 2023 de-peg without an expected-loss floor. Galati and Capalbo [18] document the SVB link earliest, Diop, Chevallier, and Sanhaji [15] rank USDC as most exposed in a daily cross-coin panel, Cintra and Holloway [8] flag the de-peg about five hours before USDC crosses \$0.99, Li et al. [25] frame it as settlement friction, and Sankaewtong et al. [35] reconstruct the on-chain contagion pathway at the transaction level. Our bound derives an expected loss from disclosed exposure, and none of these papers does.

Contribution boundary. Our work complements both lines of this literature. Structural calibrations [29] estimate a run *probability*, and event studies [3] document *flows*. We instead bound the *realized non-fundamental share* of one episode from an accounting identity. Two papers appear to conflict with this reading. Anadu et al.’s [3] *rational* run concerns the redemption decision. It does not concern whether the price was justified. Watsky et al.’s [42] weekend discount is the microstructure friction the non-fundamental share includes. Ahmed, Al-dasoro, and Duley [1] run synthetic-control tests on four *other* episodes. They treat March 2023 only as a descriptive price path. Two further papers address adjacent questions. Zhu [43] retains both large-sale and collateral risk, splitting the space of fundamental states into regions by dominant risk type, which classifies *which* channel moves price and leaves *how far* unmeasured. Molinet and Filos-Ratsikas [32] define fiat-backed coins as the degenerate always-stable case ($1/p_t = 1$), so they never analyze March 2023.

3 Setting and Data

3.1 USDC and the SVB shock

USDC is a fiat-backed stablecoin issued by Circle, redeemable one-for-one against reserves. Reserves are held in short-dated U.S. Treasuries and bank cash, so a bank failure impairs backing through the cash leg. The peg depends on full reserve backing and on an arbitrage loop: when the secondary price falls below par, an authorized participant can buy cheap tokens and redeem them at \$1.

On March 10, 2023, Silicon Valley Bank failed. Late that evening, Circle disclosed on its official X account that “\$3.3 billion of the ~\$40 billion of USDC reserves remain at SVB” [9]. Circle restated that figure as “about 8%” of total reserves in its March 12 press release [10], and contemporaneous reporting corroborates it [6]. We take $\phi \approx 0.08$, Circle’s own rounded disclosure. The raw ratio $3.3/40.0 = 8.25\%$ is a different reading of the same disclosure, and Section 4 shows the result is insensitive to the exact denominator. The secondary-market

price fell below par over March 10–11, reaching a trough of \$0.877. On March 12 at 22:15 UTC, the FDIC, the Federal Reserve, and the Treasury issued a joint statement guaranteeing SVB depositors in full [5]. USDC restored its peg over the following days. The disclosed fraction $\phi \approx 0.08$ sets the worst-case floor, and the full re-peg reveals ex post that no permanent loss occurred.

3.2 DAI and USDT

DAI is a decentralized stablecoin issued by the MakerDAO protocol. By 2023 DAI’s collateral pool was linked to USDC through its peg-stability module (PSM). The PSM is a contract that lets anyone swap USDC for DAI and DAI for USDC one-for-one, up to a debt ceiling. Press reporting says that ceiling was reached on 11 March 2023 [12]. A USDC discount can therefore be arbitrated into DAI only while the outstanding balance is below the ceiling. DAI serves as the *contagion* coin in our placebo. It is exposed to the shock through a collateral and arbitrage channel (Appendix B). DAI’s on-chain USDC backing at the run-date block is unavailable (Appendix B), so the exposure below rests on cited reporting.

Tether (USDT), the largest fiat-backed stablecoin, had no disclosed exposure to SVB and serves as the *unexposed* coin. During the run it drew flight-to-safety demand and traded at a premium above par. We exploit that upward divergence as an identifying placebo (Section 5).

3.3 Data and the frozen snapshot

All analysis runs offline on a frozen snapshot over the window 2023-03-08 to 2023-03-17 (UTC). No proprietary data enters the analysis.

Two timestamp conventions appear in the frozen series. Kraken and Coinbase label each hourly bar by its start, so 07:00 UTC means the bar covering 07:00 to 08:00. The aggregator composite instead carries the aggregator’s own observation timestamps, which fall a minute or two off the hour, so a composite reading appears as 06:59 or 07:57. Every UTC time in this paper is reported as the source recorded it, with no rounding to a common grid, so that every printed time matches a frozen value.

The headline USD price series is an hourly composite from the public aggregator DeFiLlama. The frozen record does not enumerate its constituent venues, so we describe it as an aggregator composite. We do not treat it as an independent venue. Kraken’s fiat-settled USDC/USD pair cross-checks the \$0.877 trough, with 240 complete hourly rows over 8 to 17 March, a span that contains the 216-hour analysis window. Binance USDCUSDT is a second cross-check. The frozen Binance series starts only at 14:00 UTC on March 11, after the trough, so it corroborates depth over the later hours. It cannot corroborate the trough. Because the pair is USDT-denominated, we convert it to USD by multiplying the raw price by the contemporaneous USDT-USD price, so the converted price embeds USDT’s own premium. We do not assume USDT traded at \$1. We measure

the USDT premium separately, on Coinbase USDT-USD, the second fiat-settled cross-check alongside Kraken.

To measure who redeemed, we reconstruct primary-market redemptions from on-chain burn events, FIFO-matching (first-in-first-out) each burn to the oldest unconsumed deposit that reached Circle’s minter. Appendix A reports the coverage this attribution achieves. The freeze covers the attribution’s *output*, not the per-event log it was derived from (Appendix C). For the DAI decomposition we read an archival `eth_call` read of on-chain DAI supply at the run-date block (Appendix B).

4 The Impossibility Bound

The estimator is pure accounting and uses neither a run threshold nor any fitted parameter. Let ϕ be the fraction of reserves impaired and $\rho \in [0, 1]$ the expected recovery rate on the impaired assets. The fraction permanently lost is $\phi(1 - \rho)$, so the no-panic price floor, expected backing per coin, is

$$P_{\text{fund}} = 1 - \phi(1 - \rho). \quad (1)$$

4.1 Assumptions

The bound rests on three assumptions, and a fourth serves Corollary 1.

- (A1) **Reserve-only fundamental.** A coin’s fundamental value is its expected reserve backing per unit given *disclosed* reserves. The only impairment is the exposure ϕ .
- (A2) **Par redemption when solvent.** Conditional on rescue, redemption is at \$1 *within the analysis horizon*.
- (A3) **Bounded recovery.** Recovery on impaired reserves satisfies $\rho \in [0, 1]$ because the coin is fully reserved and the reserves carry no leverage, so the worst case is total permanent loss of the impaired fraction.
- (A4) **Monotone-utility valuation.** The unconstrained marginal holder values the claim as an expected utility of its payoff, for *some* monotone increasing utility and *some* probability measure over outcomes, with no ambiguity about the disclosed ϕ . This nests risk neutrality, any degree of risk aversion, and rank-dependent weighting.

The closing clause of (A4) assumes away ambiguity about the disclosed ϕ . The 12.3% margin is the principal defense against it, and the ϕ -by-feed grid of Section 6 is the second: a holder who doubted the disclosure would have to price an impairment 4.3 points above the disclosed figure with zero recovery before the trough becomes rationalizable.

4.2 Statement and proof

Under (A1)–(A3), take a pricing measure under which the impaired fraction’s loss goes *unremediated* with probability $q \in [0, 1]$. Remediation here means an issuer make-whole, a deposit-insurance recovery, or an official backstop. The expected-loss price is

$$P_{\text{fund}}(q, \rho) = 1 - q\phi(1 - \rho). \quad (2)$$

This is the standard floor-at-recovery-value bound of credit pricing: risky debt is bounded below by its recovery value under any pricing measure [30], with q a reduced-form failure intensity [17].

Proposition 1 (Rational-pricing floor and impossibility). *Under (A1)–(A3), $P_{\text{fund}}(q, \rho) \in [1 - \phi, 1]$ for all $q, \rho \in [0, 1]$. Hence any observed price $P_{\text{obs}} < 1 - \phi$ cannot be rationalized as expected loss. The unremediated-loss probability it would require, at the worst-case recovery $\rho = 0$, is*

$$q^* = \frac{1 - P_{\text{obs}}}{\phi} > 1. \quad (3)$$

Equivalently, the price is unrationalizable for any impairment $\phi < 1 - P_{\text{obs}}$.

Proof. In (2), P_{fund} is decreasing in q and increasing in ρ on $[0, 1]^2$. It is therefore minimized at $(q, \rho) = (1, 0)$, where it equals $1 - \phi$, and maximized at $q = 0$ or at $\rho = 1$, where it equals 1. Thus $P_{\text{fund}} \in [1 - \phi, 1]$. If $P_{\text{obs}} < 1 - \phi$, no $(q, \rho) \in [0, 1]^2$ solves $P_{\text{obs}} = P_{\text{fund}}(q, \rho)$. Setting $\rho = 0$ and solving $P_{\text{obs}} = 1 - q\phi$ gives $q^* = (1 - P_{\text{obs}})/\phi$, which exceeds 1 exactly when $P_{\text{obs}} < 1 - \phi$, i.e. $\phi < 1 - P_{\text{obs}}$. $\square$

The proof never restricts q to a physical probability, so it covers any risk-neutral pricing measure. Under (A1)–(A4) the floor is preference-free:

Corollary 1 (Preference-free floor for rational valuations). *Under (A1)–(A4), the payoff on a USDC claim has support in $[1 - \phi, 1]$, and for any monotone increasing utility u and any probability measure over outcomes, the certainty equivalent $u^{-1}(\mathbb{E}[u(X)])$ of a payoff lottery X with support in $[1 - \phi, 1]$ itself lies in $[1 - \phi, 1]$. No rational valuation, under any monotone increasing utility and given the support bound, falls below $1 - \phi$. A more risk-averse holder values the claim lower within the band, but never below $1 - \phi$.*

Proof. By (A2)–(A3) the payoff satisfies $X \in [1 - \phi, 1]$ in every state. Monotonicity gives $u(X) \geq u(1 - \phi)$ pointwise, hence $\mathbb{E}[u(X)] \geq u(1 - \phi)$. Applying the increasing function u^{-1} yields a certainty equivalent of at least $1 - \phi$. The upper edge is symmetric. This is first-order stochastic dominance over a bounded support and is independent of the curvature of u . $\square$

Two distinctions govern the Corollary’s interpretation. First, it bounds rational *valuations*, meaning willingness to pay, and not market-clearing *prices*. Under binding margin or capital constraints, a forced seller can rationally transact

below every unconstrained valuation. This is the classic limits-to-arbitrage mechanism [36,21]. That channel is a friction, so it lies inside the non-fundamental share s_{nf} of (5). Second, the floor is as strong as the support bound $X \in [1 - \phi, 1]$ and no stronger. *Issuer-wrapper risk* is not covered by that bound. This is an issuer insolvency that delivers less than the floor, or delivers it late, through claim-priority uncertainty, administrative cost, or a gated distribution [20]. (A1) prices that channel at zero, on three facts. 77% of reserves sat in a segregated, SEC-registered government money-market fund custodied at BNY Mellon, not issuer-balance-sheet deposits [11]. A distribution delay is a fundamental discount only through time value, and at 2023-03-10 Treasury yields (4.81–5.01%) [39] a ~ 350 -day gate would be needed to reach the trough from the floor. Wrapper costs alone would need a 4.3-point (12.3–8.0) loss on that money-market portfolio, and no segregated government-fund structure has recorded such a loss. The floor is therefore conditional on (A1)’s disclosed-impairment support, leaving only a marketability discount that scales with *ex-ante* volatility, small relative to realized crisis volatility for short-dated Treasuries and cash [27].

4.3 Application to the March 2023 USDC trough

With the disclosed impairment $\phi = 0.08$, the floor is $1 - \phi = \$0.92$. The observed trough is $P_{\text{obs}} = \$0.877$. Prose rounds the trough to three figures throughout, while every quantity computed at the trough is evaluated on the frozen four-figure print, \$0.8767, so \$0.877 and \$0.8767 denote one reading. The trough lies below the \$0.92 floor. The implied unremediated-loss probability is

$$q^* = \frac{1 - 0.877}{0.08} = 1.54 > 1. \quad (4)$$

No admissible (q, ρ) rationalizes the trough, so the de-peg is non-fundamental by impossibility, not by assumption. The timing makes the bound conservative. The trough precedes Circle’s make-whole pledge by at most 12.5 hours and precedes the federal joint statement by 38.3 hours (22:15 UTC, 12 March 2023) [5]. The pledge carries no publication timestamp, so the 12.5-hour figure is measured to its first web-archive capture, 20:27 UTC on 11 March 2023 [11]. The floor requires no backstop concept, only that the payoff cannot fall below $1 - \phi$ state by state. Neither the trough reading nor the reserve denominator moves the conclusion, because the margin between the floor and the trough is wide relative to the variation in either. At the composite trough (\$0.8767) the lower edge of (5) is 35.1%. At the deepest USDT-denominated Binance hourly low (\$0.882) it is 32.2%. At the next-lowest composite hourly print (\$0.8949) it is 23.9%. At Circle’s alternate $\sim \$42.1\text{B}$ reserve-composition total ($\phi = 7.84\%$) the lower edge is 36.4%. At the disclosed $\sim \$40\text{B}$ total ($\phi = 8\%$, the headline used throughout) it is 35.1%. At the precise $\$3.3\text{B}/\40.0B ratio ($\phi = 8.25\%$) it is 33.1%, a 7.8–8.3% band across the three denominators, and a larger reserve total only shrinks ϕ and widens the margin by which the trough is unrationalizable. Over the same three denominators the implied q^* ranges from 1.49 to 1.57, against the headline $q^* = 1.54$ at $\phi = 0.08$.

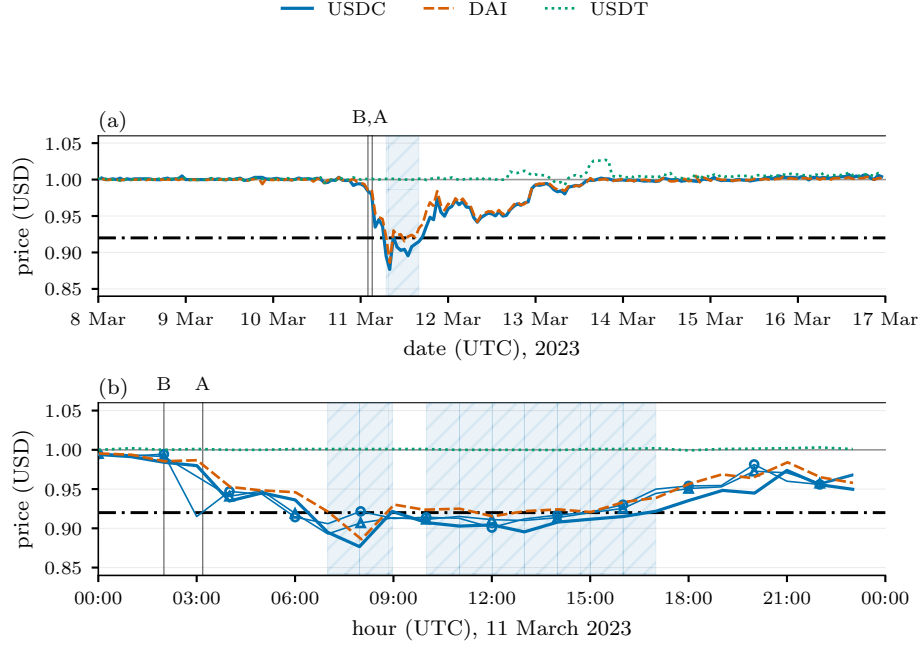


Fig. 1. Hourly USD prices of USDC (solid), DAI (dashed), and USDT (dotted) over the analysis window (a) and over 11 March 2023 (b), against par and the \$0.92 floor implied by Circle’s disclosed $\phi = 0.08$. Hatched columns in (b) are USDC’s nine breaching hours, 06:59–16:00 UTC, and the unhatched column is the recovery hour. Vertical lines mark A, Circle’s disclosure (03:11 UTC, 11 March), and B, the Coinbase USDT-USD premium peak (02:00 UTC, 11 March).

Hourly evaluation of the bound. Because the estimator requires only a price and a publicly quantified ϕ , it can be evaluated at every hour of the frozen window (Figure 1). Of USDC’s 212 in-window hourly readings, 9 lie below the \$0.92 floor. All nine fall on 11 March 2023, between 06:59 and 16:00 UTC, opening at $q^* = 1.31$ and peaking an hour later at the trough. Over the same window, DAI breaches in 2 of 214 hours and USDT in none of 213 (Table 2). *This is duration and specificity, not evidence volume:* hourly prices within one episode are near-unit-root (lag-1 autocorrelation 0.963), so the 212 readings carry an AR(1)-adjusted effective sample size of 4.0, and identification still rests on one episode and one disclosed ϕ .

4.4 The non-fundamental share

Write the observed deviation $\Delta_{\text{obs}} = 1 - P_{\text{obs}}$ and the fundamental deviation $\Delta_{\text{fund}} = \phi(1 - \rho)$. The non-fundamental share of the de-peg is

$$s_{\text{nf}}(\rho) = \frac{\Delta_{\text{obs}} - \Delta_{\text{fund}}}{\Delta_{\text{obs}}} = \frac{(1 - P_{\text{obs}}) - \phi(1 - \rho)}{1 - P_{\text{obs}}}. \quad (5)$$

Two independent restrictions give the band edges. The *lower edge* ($\rho = 0$) takes even a total permanent loss of the impaired reserves as the fundamental cap. At $\phi = 0.08$, $s_{\text{nf}}(0) = (0.1233 - 0.08)/0.1233 = 0.351 \approx 35\%$. The *upper edge* ($\rho = 1$) uses the full re-peg, which implies no permanent loss occurred, so $\Delta_{\text{fund}} = 0$ and $s_{\text{nf}}(1) = 1$: 100% ex post, an identity. This upper edge is a realized-loss identity, not an expected-loss bound. It substitutes the observed outcome, full recovery and therefore zero permanent loss, for an expectation over ρ , so it is an ex-post statement. The $\rho = 0$ lower edge is the ex-ante worst-case expectation. We report the band [35%, 100%]. The full re-peg supports the high end of that band. “Non-fundamental” means coordination panic *plus* microstructure friction, so the share is an upper bound on pure coordination panic.

Four quantities in this paper are one quantity presented four ways. q^* , $s_{\text{nf}}(0)$, the margin column of Table 3, and the floor-breach indicator are all determined by $1 - P_{\text{obs}} - \phi$, so agreement among them corroborates nothing.

4.5 Robustness to ϕ and to the assumptions

The lower edge depends on ϕ . By Proposition 1, the trough is unrationalizable for any $\phi < 1 - P_{\text{obs}} = 0.123$, i.e. for any impairment below 12.3%, a 4.3-point margin over the disclosed 8%. The non-fundamental share’s lower edge ($\rho = 0$) varies with ϕ but stays strictly positive throughout this margin, falling to $\approx 19\%$ at $\phi = 0.10$.

(A1) is the binding assumption. If a coin carried fundamental value beyond disclosed reserves, the floor would shift. For that reason ϕ is taken from Circle’s disclosure rather than fitted, and we report a band. Three further checks discipline (A1). First, the exact denominator is immaterial, as the sensitivity above shows. Second, the par-safety of the non-SVB reserves rests on Circle’s audited attestation, which itemizes them at the CUSIP level. They are $\sim \$38.5\text{B}$ of short-dated U.S. Treasuries plus cash. The attestation shows reserves exceeding USDC in circulation on both its March report dates, and the failed banks no longer appear in the March 31 schedule [13]. Third, the contagion-week worst case is bounded *ex ante*: at the trough, 77% of reserves (\$32.4B) were short-dated Treasuries in the segregated, BNY-Mellon-custodied Circle Reserve Fund, and the \$9.7B cash leg was held at several institutions, of which SVB was one [11]. Raising ϕ past 12.3% would require zero-recovery losses on more than \$5.2B of reserves, taken against Circle’s \$42.1B reserve-composition total: writing off SVB’s \$3.3B and \$1.9B of the \$6.4B cash leg held elsewhere, since the leg is \$9.7B less the \$3.3B at SVB and the threshold is \$5.2B less that same \$3.3B. No U.S. resolution on record has produced that outcome.

(A2)–(A3) are conservative: par redemption and bounded recovery both *raise* the floor, so relaxing them cannot weaken the lower edge. (A4) is dispensable for Proposition 1: it is used only by Corollary 1. Under the worst-case recovery assumption $\rho = 0$, the implied unremediated-loss probability at the composite trough exceeds the admissible $[0, 1]$ range at every ϕ in the sensitivity grid: $q^* = 1.57$ at $\phi = 0.0784$, $q^* = 1.54$ at $\phi = 0.08$, and $q^* = 1.49$ at $\phi = 0.0825$. At the opposite extreme, $\rho = 1$ collapses the pricing relation to the identity

Table 1. Cross-stablecoin placebo. “De-peg” and “premium” are the maximum deviation below and above par over the 2023-03-08–17 UTC window, and USDT’s premium appears as two rows because it differs by feed and by time. Troughs are shown to three figures and the basis-point columns are computed on the unrounded four-figure print, so the two columns are consistent at full precision rather than at the precision shown.

Coin	Exposure	Feed	Trough	De-peg	Premium (peak, UTC)	Break?
USDC	direct	composite	\$0.877	1233 bps	70 bps	yes
DAI	contagion	composite	\$0.886	1141 bps	50 bps	yes
USDT	none	composite	\$0.992	76 bps	270 bps (13 Mar 19:00 UTC) †	no
USDT	none	Coinbase	\$0.999	10 bps	266 bps (11 Mar 02:00 UTC)	no

† post-backstop: 20.8 h after the 2023-03-12 22:15 UTC joint Treasury/Fed/FDIC statement.

$P_{\text{fund}}(q, 1) = 1 - q\phi(1 - 1) \equiv 1$ for every $q \in [0, 1]$, so full assumed recovery makes the floor coincide with par and the bound uninformative. This is why Proposition 1 and Corollary 1 are evaluated at $\rho = 0$ throughout.

5 Cross-Coin Evidence

A global-games run model (Appendix E) yields an ex-ante fragility ordering for the March 2023 shock, driven by reserve exposure: USDC (direct SVB exposure) most fragile, DAI (mechanically pinned to USDC through the PSM) next, and USDT (no SVB exposure) least. This single episode identifies exposure. It does not identify concentration, since USDC was less concentrated than USDT on the external data of [29] (Appendix A) and still had the deepest trough. DAI is mechanically linked to USDC (Appendix B), so the null leaves one free comparison, USDT against the USDC–DAI pair. Under that null the realized ordering has probability 1/2, and we therefore treat the match as illustrative.

Realized ordering. Troughs are \$0.877 (USDC), \$0.886 (DAI), and \$0.992 (USDT), matching the ex-ante fragility ordering (Figure 1a, Table 1). USDC and DAI clear the 1% break threshold, a deviation of at least one cent from par, by factors of 12 and 11. That threshold is one we set, and the calm-baseline false-positive check of Appendix D tests it. USDT does not clear it, so the placebo’s criterion is satisfied, namely that the exposed coins break and the unexposed coin does not. The de-peg printed on material volume: in the frozen Binance USDCUSDT klines, the minimum hourly low (\$0.882, 2023-03-11 14:00 UTC) printed on \$109M of volume in that single hour. The series starts after the trough, so it corroborates depth on volume (Section 3). The impossibility result holds at every in-window reading, including \$0.882. DAI’s break runs through its documented USDC exposure (Appendix B).

Redemption concentration. Address-level redemption concentration is $\text{HHI} = 0.256$. The larger of the two dominant conduits is labeled Coinbase in two public

label sets, `dawsbot/eth-labels` and `brianleect/etherscan-labels`, and carries 40.9 of the 69.6 points of volume share (rounded here and given to four figures in Appendix A). The second conduit’s Coinbase attribution comes from our own on-chain tracing, with no public label. Excluding both conduits leaves the index at 0.067. The no-sybil condition, meaning no entity outside the two dominant conduits splits redemptions across multiple funding addresses, is untestable from the public label coverage reported in Appendix A. Three distinct quantities are easily conflated here. 40.9% is the largest single address’s share of captured redemption volume, and 69.6% is the two dominant conduits’ combined share. A separate figure for USDC’s largest monthly arbitrageur, reported in [29], uses a different unit and is not a replication (Appendix A).

USDT as the identifying placebo. USDT was bid to a premium while USDC and DAI fell [4]. \$1.24B of \$1.52B in window volume on Coinbase traded above par, so the Coinbase premium held on material volume. The premium is venue- and time-specific. On Coinbase USDT-USD, it peaked at +266 bps at 02:00 UTC on 11 March, 71 minutes *before* Circle’s 03:11 UTC disclosure. The premium is a response to the SVB failure itself, because Circle’s USDC-specific disclosure had not yet been made. In the 02:00 UTC hour, Kraken USDT/USD printed a +600-bps high on two trades totaling 225 USDT, against a volume-weighted average premium of +95 bps for that hour. A third peak of +270 bps appears in the composite feed at 19:00 UTC on 13 March. It follows the joint Treasury/Fed/FDIC statement of 22:15 UTC on 12 March [5] by 20.8 hours. This peak is therefore a post-backstop event and is not the flight-to-safety move the identification argument uses. The 02:00 UTC premium is a flight-to-safety signature, and no market-wide repricing of stablecoin risk can produce it, because such a repricing would move all three coins down together.

Coins excluded from the placebo set. Three further zero-SVB-exposure coins, USDP, GUSD, and BUSD, fail the 1% break test’s diagnostic requirement because their pre-SVB price feeds already print below \$0.99, and their zero-exposure flow evidence is consistent with the placebo result (Appendix D).

6 Robustness and the Specificity Panel

The headline result uses the aggregator composite, the only feed covering every hour of the window, corroborated by the fiat-settled venue that carries USDC. Kraken’s USDC/USD close breaches the \$0.92 floor in 9 of 240 hours and its VWAP series breaches it in 10 (Table 2). The hourly low breaches it in 12, but the three extra hours rest on one print: the \$0.874 low is a 3,373.52-USDC trade worth \$2,948, about 0.01% of that hour’s traded volume. Binance corroborates depth only (Section 3).

At $\phi = 0.08$, restricted to the four price feeds’ own troughs (composite, Kraken close, Kraken VWAP, and Binance/USDT-deflated), the non-fundamental share $s_{\text{nf}}(0)$ ranges from 19.2% (Kraken close) to 35.1% (composite). A broader sensitivity range should not be conflated with it. That range

Table 2. Breaching hours against the \$0.92 floor, over each series’ own observed window, for the feeds that cover the trough: the composite for USDC, DAI, and USDT, and Kraken’s close, VWAP, and low for USDC. Binance is excluded because its series begins after the trough.

Feed	Breaches
Composite (USDC)	9/212
Kraken close (USDC)	9/240
Kraken VWAP (USDC)	10/240
Kraken low (USDC)	12/240
DAI (composite)	2/214
USDT (composite)	0/213

runs over the full grid of three ϕ conventions (0.0784, 0.08, 0.0825) and all five trough readings, which add the composite feed’s second-lowest print (\$0.8949) to the four feed troughs above. It spans 16.7% to 36.4%. Kraken’s close series is autocorrelated (lag-1 = 0.945), so its 240 readings carry an effective sample size of 6.7, and the 9 breaching hours measure a duration rather than 9 statistically independent observations.

The impairment loss is fixed in dollars while redemptions shrink the reserve base, so ϕ moves inversely with it. At the composite trough ϕ is 0.079, below the headline 0.08, so the floor rises to \$0.921. Supply is observed only daily. The two snapshots bracketing the trough admit an interpolation band $\phi \in [0.078, 0.081]$, over which the count of breaching hours ranges from 9 to 11. Computed hour by hour, ϕ reaches its highest value in the last hour of the window, and that maximum is a censoring artifact. At that hour the coin trades at \$1.004 and $q^* = -0.044$, so the bound is vacuous there and the maximum falls on the last observed day. We report the band as a bound on how far the correction can move ϕ . Daily supply data cannot resolve a finer measurement.

Why only one episode meets the preconditions. The estimator requires three preconditions. First (i), ϕ must be publicly quantified while the exposure is unresolved. Second (ii), the price must fall more than 1% below par while that disclosure is live. Third (iii), the fundamental channel must be reserve-only, so that $1 - \phi$ is the floor. We screened seven candidates: BUSD, Tether (2022), DAI (2020), USDR, USDP, GUSD, and TUSD. *None satisfies all three.* Each candidate fails in one of two ways: no disclosed impairment or a loss measured under a different valuation model (Appendix D). Issuers with a material impairment did not disclose its size in any of the seven screened cases, so one qualifying episode is a property of the disclosure record.

Tether’s episode is assessed under precondition (i): ϕ publicly quantified while the exposure was unresolved. The NYAG’s 25 April 2019 statement quantified Tether’s exposure and Tether contested the characterization. The statement implies $\phi \in [0.259, 0.332]$. The deepest post-disclosure Kraken USDT/USD print, \$0.9551 on 26 April 2019, is 21.4 to 28.8 cents above every worst-case floor.

Table 3. Specificity panel across three candidate episodes. USDC (2023) is the panel’s one true positive, with the floor breached under all sixteen combinations of four price conventions and four ϕ readings: the three disclosed denominators of Section 4 and the dilution-adjusted $\phi = 0.079$. The row shown is $\phi = 0.08$ against the composite trough, and q^* is undefined where $\phi = 0$.

Episode	Disclosure	ϕ	Floor	Trough (feed)	q^*	Fires	Margin
USDC Mar 2023	[9], 11 Mar 03:11 UTC	0.08	0.92	composite, headline: \$0.877 (11 Mar 07:57 UTC)	1.54	Yes	−0.043
Tether 2019 [a]	NYAG press release, 25 Apr 2019	[0.259, 0.332]	[0.668, 0.742]	Kraken, low: [0.135, \$0.955 (26 Apr 02:00 2019 02:00 UTC)	[0.174]	No	[0.214, 0.288]
USDT, Terra episode (May 2022), not counted [b]	no disclosure	0.00	1.00	composite: \$0.993	–	Yes	−0.007

[a] regulator-quantified, issuer-contested. [b] degenerate by construction, with $\phi = 0$, and not counted as a panel member.

USDT, Terra episode (May 2022), is the degenerate control, with $\phi = 0$, a floor at par, and q^* undefined. Paxos’s USDP is a timing control and is not in the panel. Its only quantified exposure was \$250M at Signature Bank. Paxos disclosed it 80 minutes after the joint statement [5] had already guaranteed the deposits Paxos described.

The panel holds one true positive, one episode on which the bound returns no verdict, and one degenerate control (Figure 2). USDC in 2023 is the true positive: the floor is breached under all sixteen combinations of ϕ and price convention, and the trough requires an unremediated-loss probability of 1.20–1.61. Tether in 2019 is regulator-quantified and issuer-contested, with $\phi \in [0.259, 0.332]$, a trough 21.4–28.8 cents above its floor, and $q^* \in [0.135, 0.174]$. The degenerate control is USDT, Terra episode (May 2022), with $\phi = 0$, not counted. We exclude the Reserve Primary Fund of 2008 because its published figure is a NAV rather than a market price. The Tether row’s ϕ comes from a regulator, who quantified the exposure publicly at the time, and not from Tether’s own disclosure. Because the quantification is the regulator’s and Tether contests it, the panel contains no validated true negative under precondition (i), ϕ publicly quantified while the exposure was unresolved. The specificity of the bound is unmeasured.

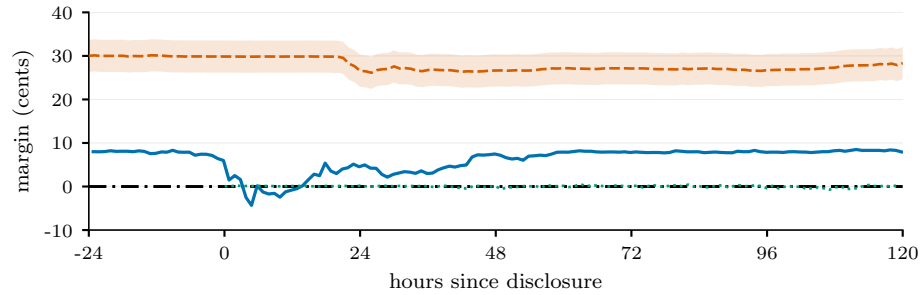


Fig. 2. Each episode’s margin, price minus its own floor in cents, from 24 hours before to 120 hours after its own quantification. Zero is the floor: USDC 2023 (solid) breaches it, Tether 2019 (dashed, with its regulator-quantified floor range shown as a band) and USDT, Terra episode (May 2022) (dotted), do not. The Terra floor is par because $\phi = 0$.

7 Policy and Limitations

Policy implications. An 8% *disclosed* bank exposure produced a de-peg at least 35% non-fundamental by the bound and exactly 100% ex post, an identity, reversed once the Federal Reserve, Treasury, and FDIC guaranteed SVB deposits [5]. The episode is an argument for limiting idiosyncratic bank-deposit exposure.

The size of microstructure friction. The time-value discount on a redemption delayed two to three days by banking hours [42] is 2.6–4.1 bps [39]. That is more than two orders of magnitude short of the 1,233-bps gap. The one remaining channel is limits-to-arbitrage with redemptions suspended [36,21], which the residual already contains (Section 4).

Limitations. The panic-versus-fundamentals split is contested [3,42]. The composite feed could mismeasure the trough. We address this with the Kraken cross-check, the venue-level USDT series, and the material-volume Binance print at \$0.882, which corroborates the depth of the move rather than the trough itself (Sections 6 and 5). The disclosed ϕ could understate exposure, but the 12.3% margin and the ϕ -by-feed grid show the conclusion survives. The non-fundamental share bundles panic with microstructure friction, so we claim only an upper bound on pure coordination panic. DAI’s USDC exposure is reporting-based. MakerDAO’s PSM USDC was off-chain by March 2023, so no on-chain series exists (Appendix B). We measure the redemption HHI at the address level, so it is an upper bound on entity-level concentration (Appendix A). The no-sybil condition that this upper bound requires is untestable with public labels, which resolve only 2 of the 776 funding addresses. We identify the mechanism from a single episode. The ex-ante fragility ordering identifies *exposure* alone, and under the null of Section 5 the realized ordering carries probability 1/2. Within the

episode, hourly readings are autocorrelated, giving an effective sample size of 4.0. The ex-post 100% is an aggregate statement: holders who sold at the trough still realized losses.

Scope condition: ϕ must be public. TrueUSD had an unremediated reserve impairment during our window, but nothing about it was publicly quantified until a September 2024 SEC complaint [40]. On 11 March 2023, TUSD traded 116 bps below par while USDC traded 1,233 bps below. Our estimator returns null on TUSD (Appendix D). We define the residual *relative to what the market could know*: concealed impairment produces no detectable signal by construction. The scope condition is an argument for the quantified reporting regimes the GENIUS Act contemplates [38].

8 Conclusion

A disclosed impairment implies a worst-case floor. The March 2023 USDC trough lies below it, so at least 35% of the de-peg was non-fundamental by impossibility, and exactly 100% ex post, an identity. Against a small panel, the instrument returns one true positive, and its specificity is unmeasured (Section 6). Two policy readings follow (Section 7): limit idiosyncratic bank-deposit exposure, and require the quantified disclosure the instrument depends on. The instrument requires no structural model, only a price and a publicly quantified ϕ .

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A Redemption Concentration

We measure redemption concentration on-chain (Section 3) to ask whether a few large arbitrageurs caused the run. We attribute USDC burns to the external deposits that funded them by first-in-first-out matching, so every counted dollar corresponds to a burn. This attribution covers 89.8% of the \$8.15B gross burned. The remaining 10.2% is the pre-window remainder, burns whose funding deposit arrived before the frozen window opens. Over the 776 funding addresses, Table 4 reports both the raw index, $\text{HHI} = \sum_i s_i^2$ over redemption-volume shares s_i , and a size-corrected normalized variant, $(\text{HHI} - 1/N)/(1 - 1/N)$, which removes the floor a finite N imposes. At $N = 776$, the two differ by under 0.001 (0.2558 raw vs. 0.2548 normalized). We therefore report the raw index throughout, and an unqualified “HHI” means raw. Redemption outflow accompanied the de-peg: USDC’s circulating supply fell from \$43.29B on 7 March to \$35.90B on 18 March, a *net* decline of \$7.39B, or 17.1%, over the window. The \$8.15B above is *gross* burns, which exceed the net decline because minting continued through the window.

Upper bound on ultimate-redeemer concentration. The two largest funding addresses (69.6% of captured volume) are single-source routing conduits that are both funded by one Coinbase hot wallet, which itself aggregates 14,111 distinct senders. At the presenting-entity level, Coinbase’s 69.6% implies an entity HHI near 0.49. That concentration measures a custodian’s pass-through volume, but the claim here concerns *ultimate redeemers*. Disaggregating a custodian strictly lowers an HHI, so the address-level figure is an upper bound on ultimate-redeemer concentration *under a no-sybil condition*: no entity outside the two dominant conduits splits its burns across multiple funding addresses [41].

Table 4. Redemption concentration, with an entity-resolved block. The top block reports address-level figures, and the entity-resolved block joins the 776 funding addresses against two public label sets (dawsbot/eth-labels, brianleect/etherscan-labels), which together resolve 2 of 776 addresses: the larger conduit (40.9% of volume) carries an external Coinbase label, and the smaller (28.6%) does not, so its Coinbase attribution rests on the paper’s own on-chain tracing. At this coverage no entity holds two or more of the 776 addresses, so entity HHI equals address HHI to full precision, which means the label evidence is too sparse to test the no-sybil condition.

Statistic	Value
Wallets (redeemers)	776
Coverage of gross burns	89.8%
HHI (raw)	0.2558
HHI (normalized)	0.2548
Top-10 share	88.1%
Largest share	40.9%
Effective # redeemers (1/HHI)	3.9
Bootstrap 95% CI (HHI)	[0.0517, 0.4184]
HHI, ex-custodian	0.0671
Largest share, ex-custodian	16.8%
Effective # redeemers, ex-custodian	14.9
<i>Entity-resolved (public label join)</i>	
Labelled coverage	2/776
Externally corroborated share	40.94%
Own-tracing share	28.63%
Entity HHI (0 multi-address entities)	0.2558
No-sybil condition	untestable with public labels
HHI, labelled custodians excluded	0.2528
HHI, both Coinbase conduits excluded	0.0671
Supremum worst case	0.9966

Two public label sets (dawsbot/eth-labels and brianleect/etherscan-labels) resolve only 2 of the 776 funding addresses, so the no-sybil condition remains *untestable*. At that coverage, the join would detect a two-address entity with probability 3.33×10^{-6} . A null result from that join is therefore uninformative. The larger conduit (40.94% of volume) carries the external label “Coinbase: Circle Deposit 1”. The smaller (28.63%) is unlabeled in both sets, and its Coinbase attribution comes from our own on-chain tracing (Section 3). The two sum to 69.57%, which the prose rounds to 69.6% throughout. Excluding only the externally labeled custodian leaves the index at 0.2528, and the drop to 0.0671 requires excluding the self-attributed, unlabeled conduit as well. We therefore report 0.0671 as conditional on our own attribution, and 0.2528 as the label-only floor. No informative worst-case cap follows from the label evidence: at that coverage, the label-consistent supremum

is 0.9966. That supremum merges the entire unlabeled tail with the one labeled custodian, so only our own attribution bounds the entity HHI.

Ma–Zeng–Zhang [29] report 521 redeeming arbitrageurs per month for USDC against 6 for USDT, with USDC’s largest single arbitrageur executing 45% of monthly redemptions against USDT’s 66%. That is a different dataset and a different unit from our 40.9% and 69.6% address-volume shares, so it does not replicate them. The external label covers 40.9 of the 69.6 points of volume share. Within that share, redemption ran through custodial conduits rather than through a few large arbitrageurs. Beyond it, the same conclusion follows from our own attribution, with no external evidence behind it.

B DAI’s Documented USDC Exposure

DAI broke (\$0.886) through a contemporaneously documented exposure to USDC. The break was not a separate coordination run, and counting it as one would double-count the USDC run, so this appendix does not decompose DAI’s 1,141-bps de-peg into mechanical and panic components.

DAI was linked to USDC through MakerDAO’s peg-stability module (PSM), which swaps USDC $\leftrightarrow$ DAI one-for-one up to a debt ceiling, and through USDC held as vault collateral. A press report published at 10:10 UTC on 11 March 2023 and web-archived 44 minutes later gives two distinct ratios in a single passage [12]. DAI was “about 36% backed by USDC, according to daistats” and the USDC PSM had “reached its cap of 3.1 billion USDC, so no more DAI can be minted with USDC”. Against the on-chain DAI supply at 2023-03-11 12:00 UTC (\$4.19B), the \$3.1B cap corresponds to 74% of DAI outstanding having been minted through the USDC PSM. The 74% is a different quantity from the source’s own 36% figure, which is a share of total collateral rather than of DAI supply. We report both ratios (Table 5) and use neither as a pass-through coefficient, because the two denominators are not interchangeable.

Three facts prevent a decomposition. First, the two ratios differ by roughly a factor of two and imply mechanical components of 444 and 912 bps against the same observed 1,141 bps, so any residual would be an artifact of which ratio we choose rather than a measurement. Second, the source sentence that reports the \$3.1B also records that the USDC $\rightarrow$ DAI mint leg had reached its cap at the moment we analyze. That leg’s arbitrage would move DAI’s price toward USDC’s, so we cannot argue that the PSM raises effective pass-through toward 100%. Consistent with the cap being binding, DAI traded at or above USDC through the de-peg rather than converging to it. Third, we have no on-chain custody reading to compare with the reported figure. The PSM’s USDC had moved to off-chain custody by 2023, so the module and its collateral adapter both hold approximately zero USDC at the run-date block, and the \$3.1B stands as a reported debt-ceiling figure.

DAI carried a documented USDC exposure that USDT did not. DAI broke and USDT did not. DAI’s break is therefore not independent evidence of a second

Table 5. DAI’s de-peg and its documented USDC exposure. We report both ratios the single contemporaneous source gives, 74% of DAI supply minted through the USDC PSM at its \$3.1B cap and the source’s own 36% USDC share of DAI collateral, because they have different denominators and imply materially different pass-through. We do not select between them and report no mechanical/panic split. The \$3.1B is a cited reporting figure (The Block, 11 March 2023) with no on-chain corroboration, for the reason given in Appendix B.

Quantity	Value
USDC de-peg (trough, composite)	1233 bps
Observed DAI de-peg (trough, composite)	1141 bps
On-chain DAI supply, run-date block	\$4.19B
Reported PSM-USDC-A cap, 11 Mar 2023	\$3.1B
as a share of DAI supply	74%
Reported USDC share of DAI collateral	36%

coordination run. We make no claim about how much of DAI’s 1,141 bps was mechanical.

C Reproducibility

All results derive from a frozen snapshot cataloged in Table 6, whose source URLs come from the data manifest, which also records a SHA-256 digest for every file it lists. Every stochastic routine is seeded ($\text{SEED} = 20230311$).

The hourly coverage is not uniform across coins. USDC is missing 4 of 216 expected hourly points (17:00 UTC on 13 March, 14:00 UTC on 14 March, 17:00 UTC on 14 March, and 12:00 UTC on 15 March), DAI 2 of 216 (17:00 UTC on 13 March and 16:00 UTC on 14 March), and USDT 3 of 216 (20:00 UTC on 11 March, 17:00 UTC on 13 March, and 17:00 UTC on 14 March). One hour, 17:00 UTC on 13 March, is missing from all three.

Every reported number and figure regenerates from this snapshot via the following command sequence: `python data_fetch.py verify` $\rightarrow$ `python data_fetch.n1.py verify` $\rightarrow$ `python redemptions.py --from-frozen` $\rightarrow$ `python -m pytest` $\rightarrow$ `python report.py`. None of these five steps makes a network call or reads `ETHERSCAN_API_KEY`, and none can overwrite a file already recorded in the data manifest. `make fc27` also pins `SOURCE_DATE_EPOCH` to a fixed constant and the PDF trailer /ID via `\pdftrailerid{}`, so the typeset PDF is byte-reproducible. Tables 7 and 8 name the file and key behind each.

Acquisition is a separate step and requires network access. Six scripts wrote the frozen snapshot. `data_fetch.py`, `data_fetch.n1.py`, `second_event.py`, and `usdt_premium.py` are keyless. `redemptions.py` and `dai_contagion.py` also require an Etherscan key. The key is read from the environment only and is never written to any artifact. Re-running any of the six against live data contacts an external API and overwrites the corresponding frozen file. The reproduction sequence above does not require this step and should not begin with it.

Table 6. Frozen data snapshot. Every row’s count is read directly from the file itself. The 212/213/214-point spread across USDC/DAI/USDT in row 1 reflects independent gaps in the upstream aggregator’s per-coin coverage, enumerated in the text above.

Series	Source	Rows	Coverage
USDC/USDT/DAI price	DeFiLlama	639	7–16 Mar 2023
USDC/USDT cross	Binance	131	11–17 Mar 2023
USDC supply	DeFiLlama	12	7–18 Mar 2023
Redemptions	Etherscan	776	8–17 Mar 2023
DAI backing	Etherscan	1	11 Mar 2023
USDT/USD price+vol	Coinbase	217	8–17 Mar 2023
Terra price	DeFiLlama	384	7–15 May 2022
Kraken USDC/USD	Kraken	240	8–17 Mar 2023
Kraken USDT/USD	Kraken	240	8–17 Mar 2023
Kraken USDT/USD 2019	Kraken	624	20 Apr–15 May
NYAG press release	NYAG	4	25 Apr 2019
Paxos/USDP state-ment	Paxos	3+12+1	8–14 Mar 2023
Labels (dawsbot)	GitHub	144,378	–
Labels (brianleect)	GitHub	23,288	–
Block anchors	Ethereum RPC	240	8–17 Mar 2023

The FIFO burn attribution underlying `data/redemptions_by_wallet.csv` is not reproducible offline. We freeze and hash the attribution’s *output*, a 776-row wallet-to-volume table. We do not freeze its input, the raw on-chain transfer logs the attribution FIFO-matches. `redemptions.py --from-frozen` re-validates that frozen output against its own manifest record. It does not repeat the attribution. The \$3.1B PSM figure (Appendix B) and the two public label sets (Appendix A) are reporting-based in the same way: we freeze and hash what they state, and the pipeline does not recompute them.

An anonymized artifact repository is available at <https://anonymous.open.science/r/usdc-depeg-artifact-F62D/>.

D Excluded Coins and Candidate Episodes

Three further zero-SVB-exposure coins exist: USDP, GUSD, and BUSD. The exclusion criterion is calm-window behavior alone. Over a calm, pre-SVB baseline (February 16–March 8, 2023), USDP, GUSD, and BUSD already print below \$0.99 in 7, 1, and 1 of 480 hours. The 1% break test therefore has no discriminating power for them, and we exclude them before reading any crisis-window price. For completeness, USDP, GUSD, and BUSD trough at \$0.948, \$0.928, and \$0.985. Liquidity offers no alternative exclusion rule. Frozen median daily vol-

umes range from $\sim \$16\text{M}$ for USDP, which was listed on Binance mid-window, to $\sim \$687\text{M}$ for BUSD. GUSD has no freezable series. BUSD is further confounded by an unrelated regulatory event, the February 2023 NYDFS order halting new issuance. Anadu et al. [3] find GUSD and USDP had no significant dollar flows. Their finding corroborates independently that the run was coin-specific. Three placebo coins whose feeds can register a 1% break support the test better than a six-coin panel in which three cannot.

The 1% break threshold and the worst-case floor are distinct: a coin can cross the 1% materiality threshold without nearing the price at which no reserve loss could rationalize the price. The bound is evaluated at each hourly price, so breaching hours need not be contiguous. USDC’s impossibility window is interrupted at 08:57 by a recovery to $\$0.9216$, a price 16 bps above the floor, at which $q^* = 0.98$, before falling back below the floor for seven further hours. USDT’s worst in-window hour reaches $q^* = 0.095$, an order of magnitude below the $q^* = 1$ firing threshold.

TrueUSD and the public-information scope condition. The scope condition of Section 7 rests on a documented case. TrueUSD is excluded on timing: its impairment was quantified publicly only after our window closed. The SEC’s September 2024 complaint alleges that a substantial fraction of TrueUSD’s backing sat in an offshore commodity fund through our window [40]. The complaint became public eighteen months after the March 2023 episode had ended. Precondition (i) requires public quantification while the exposure is unresolved, and TUSD had none. TrueUSD’s contemporaneous attestations used unquantified “cash and cash equivalents” language and never named the fund. Run on TUSD in March 2023, the estimator would have drawn $\phi = 0$ from those attestations and correctly returned null despite a then-unquantified impairment. TUSD therefore illustrates the scope condition and enters no test, because its de-peg depth over 2022–2024 is uncorrelated with its alleged impairment trajectory.

The seven candidate episodes. BUSD and Tether (2022) are genuine price dislocations with no disclosed own-reserve impairment. Paxos affirmed full 1:1 backing throughout while redeeming $\sim \$7.9\text{B}$ of BUSD in a month [34], and Tether’s disclosures give asset-composition breakdowns and denials of rumors without stating an impairment fraction. Run on either, the estimator sets $\phi = 0$ and reproduces the degenerate Terra null (Section 6). Neither gives a second identification. DAI (2020, “Black Thursday”) [22] has a disclosed bad-debt figure but is a collateral-liquidation cascade requiring a different fundamental-value model, violating precondition (iii). USDR is a tokenized-real-estate hybrid violating (A1). USDP and GUSD fail the diagnostic-print test above. TUSD fails precondition (i).

E The Run Mechanism

The model below makes the run mechanism and the non-fundamental residual’s friction term explicit, and it yields an ex-ante cross-coin prediction. The model

does not produce the reported numbers. Section 4's estimator derives those by accounting alone.

A unit mass of holders choose REDEEM or HOLD, with the HOLD payoff normalized to 0 in both states, so payoffs below are *net gains of REDEEM over HOLD*. If the peg holds, redeeming forfeits the holding benefit and pays the transaction cost, a net gain

$$g_n = -(\eta + c) < 0, \quad (6)$$

where $\eta \geq 0$ is the forfeited yield and liquidity premium and $c \geq 0$ the redemption cost. If the peg breaks, a holder who redeems early avoids the reserve haircut, a net gain $g_b > 0$. The realized liquid buffer is θ , with $\mathbb{E}[\theta] \approx 1 - \phi$, and a run succeeds iff the redeeming mass n reaches θ .

By the global-games selection result [7,33,19], the model admits a unique threshold equilibrium: the marginal holder at θ^* holds a uniform belief over the redemption mass, $n \sim U[0, 1]$, so $\Pr(\text{break}) = \Pr(n \geq \theta^*) = 1 - \theta^*$. Setting the REDEEM and HOLD payoffs equal gives $(1 - \theta^*)g_b + \theta^*g_n = 0$. With $g_n = -(\eta + c)$ this solves to

$$\theta^* = \frac{g_b}{g_b + \eta + c}. \quad (7)$$

A self-fulfilling run occurs when $\theta < \theta^*$, so a larger θ^* means a *more* fragile coin.

The gain g_b increases in a coin's reserve *exposure* to the shock, because more impaired backing means a larger haircut to avoid. By (7), higher exposure implies a higher θ^* and greater fragility. This mechanism gives the ex-ante fragility ordering applied in Section 5.

Table 7. Claim map, first half. Each reported quantity, the results file it regenerates from, and the key within that file. Both the file and the key are checked against the results files themselves, and the section column is a cross-reference rather than a typed number, so neither can drift.

Claim	Sec.	Results file	Key
Worst-case floor and implied unremediated-loss probability	4	number1b_rational_bound.json	p_obs, floor_worstcase, implied_failure_prob_rho0
Non-fundamental share, lower edge and band	4	number1_nonfundamental_share.json	headline_band.low, headline_band.high
Denominator sensitivity of phi and the share	4	phi_denominator_sensitivity.json	phi_conventions, grid, feed_trough_only_at_phi_0.08
Dated impossibility window and per-coin firing counts	4	impossibility_window.json	impossibility_region_usdc, coins.USDC.independence.effective_n
Dilution-adjusted phi(t) and its interpolation band	6	phi_dynamic.json	variant_i, interpolation_band, firing_set
Cross-feed trough corroboration and below-floor depth	6	trough_corroboration.json	firing_report, below_floor_depth, independence
Hourly observation counts per coin	C	hourly_counts.json	per_coin, window.expected_hours
Cross-coin placebo troughs and break threshold	5	number3_placebo.json	USDC, DAI, USDT, break_threshold
USDT premium depth and timing	5	usdt_premium.json	max_premium_bps, premium_volume_usdt
USDT premium peak timestamps by venue	5	usdt_premium_timing.json	peaks.coinbase_usdt_usd, peaks.composite_usdt
Redemption concentration and entity resolution	A	number2_redemption_hhi.json	upper_bound.hhi_raw, entity_resolved, coverage

Table 8. Claim map, second half. Columns as in Table 7.

Claim	Sec.	Results file	Key
FIFO burn-attribution rule	A	fifo_rule.json	matching_window, tie_break, truncation
DAI reported USDC exposure	B	number4_dai_contagion.json	psm_share_of_dai_supply, source_collateral_share, no_decomposition
Specificity panel rows and breach combinations	6	specificity_panel.json	row1_usdc_march_2023, row2_tether_2019, row4_terra_2022
Terra degenerate control	6	second_event_terra.json	phi_terra_exposure, degenerate_caveat
Excluded-coin calm-baseline evidence	D	placebo_excluded_coins.json	USDP, GUSD, BUSD
Exposure-ordering null probability	5	ordering_null.json	probability_as_fraction
TrueUSD scope-condition deviation	7	tusd_2023.json	deviation_from_par_bps
Make-whole pledge timing bound	4	makewhole_timing.json	circle_makewhole_pledge_by_trough_reference
Time-value discount on a delayed redemption	7	time_value.json	discount_bps_range_2to3d
Trough print volume	5	trough_volume.json	volume_in_trough_hour_usdc
Hour-by-hour panel series for 11 March	4	panel_11mar.json	panel, cross_coin_cross_window_summary